

Design and Analysis of First-Order Active Filters



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ETE 3101 Circuit Analysis II Laboratory

Lab 2

March 8, 2026

TABLE OF CONTENTS

1. ABSTRACT.....	[Page #3]
2. INTRODUCTION	[Page #3]
3. REVIEW.....	[Page #4]
3a. Conceptual analysis.....	[Page #4]
3b. Active filter transfer function chart	[Page #5]
3c. Passive transfer function chart	[Page #7]
4.METHODOLOGY.....	[Page #9]
5. CIRCUITS.....	[Page#11]
5a.Active Inverting Low Pass	[Page #11]
5a1. Simulated circuit and data.....	[Page #11]
5a2.Calculations, analysis, and results	[Page# 12]
5b. Active Inverting High Pass.....	[Page#19]
5b1. Simulated circuit and data.....	[Page #19]
5b2.Calculations, analysis, and results	[Page#20]
5c.Active Non Inverting Low Pass.....	[Page #26]
5c1. Simulated circuit and data.....	[Page #26]
5c2. Hardware Implementation.....	[Page #27]
5c3.Calculations, analysis, and results.....	[Page#27]
5d.Active Non-Inverting High Pass	[Page #35]
5d1. Simulated circuit and data.....	[Page #35]
5d2. Hardware Implementation.....	[Page #36]
5d2.Calculations, analysis, and results	[Page #36]
6. DISCUSSION AND CONCLUSION	[Page #43]
7. REFERENCES	[Page #44]

ABSTRACT

This paper investigates the behavior and design of first-order active filters using operational amplifiers and passive RC networks. Four configurations were analyzed: non-inverting low-pass, inverting low-pass, non-inverting high-pass, and inverting high-pass. Transfer functions were derived analytically and verified using Multisim simulations. Selected circuits were constructed on a breadboard and measured experimentally. Theoretical cutoff frequencies and -3 dB points closely matched simulated and measured results, confirming expected filter behavior and stability characteristics.

INTRODUCTION

Filters are circuits designed to selectively allow signals within a desired frequency range while attenuating unwanted frequency components. They are commonly used in signal processing, communication systems, and instrumentation. Filters can be implemented as passive filters (as seen in lab 1) and active filters (as seen in this lab), as well as composite (combinations of low, high, and bandpass filters).

Electronic components can be classified as passive or active devices. Passive devices include resistors, capacitors, inductors, and diodes. These components do not provide a gain in power—they just store, dissipate, or direct energy. Active devices include transistors and op amps (operational amplifiers). These components require an external power source and are capable of providing amplification. When passive elements (such as an RC filter circuit) are combined with active elements (such as an op amp), this results in an active circuit.

An active filter circuit consists of an op amp configured in either an inverting or non-inverting configuration that provides gain, along with a passive RC network that determines

the filtering behavior of the circuit. Active filters also allow greater control over the attenuation of frequencies beyond the cutoff frequency and provide several advantages including voltage gain. In this lab, the behavior of active filter circuits is analyzed by examining their bode plots, transfer functions and frequency responses in order to determine the filtering characteristics of the circuits.

REVIEW

Filters can also be classified by their order. The order of a filter corresponds to the highest power of S in the denominator of its transfer function. A first-order circuit contains resistors, capacitors, *or* inductors. A second-order circuit contains two first-order circuits. Similarly, a third-order circuit consists of a second-order circuit plus a first-order circuit. Increasing the order of a filter produces a steeper attenuation beyond the cutoff frequency, which gets the circuit closer to the ideal goal of reducing unwanted frequency components outside of the desired bandwidth. In this lab, the four filter circuits built are first order.

To analyze filter behavior, the transfer function of the circuit is determined. For active filters, the transfer function consists of the gain of the op amp multiplied by the transfer function of the passive RC network. The gain of the op amp depends on whether the amplifier is configured as inverting or non inverting. For non inverting active filters, the input signal is applied to the non inverting (+) terminal of the op amp and the inverting (-) terminal is connected to ground. The output signal remains in phase with the input signal, hence the waveform is not inverted. Its gain is determined by the feedback resistor, and is $A_v = 1 + \frac{R_f}{R_{in}}$. In an inverting amplifier, the input signal is applied to the inverting (-) terminal of the op amp through an input resistor, while the non-inverting terminal is connected to ground. The gain of an inverting

amplifier is determined by $Av = -\frac{R_f}{R_{in}}$. In all four configurations, feedback resistors (R_f) are present. A buffer occurs when the gain of the amplifier is equal to one. In non-inverting configurations, this can be achieved when the feedback resistor R_f is set to 0, resulting in $Av = 1 + \frac{0}{R_{in}} = 1$.

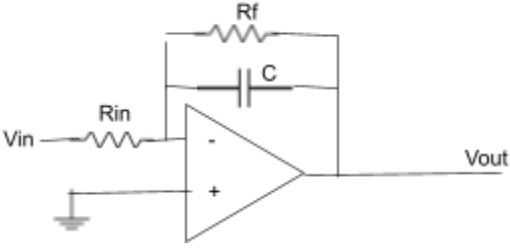
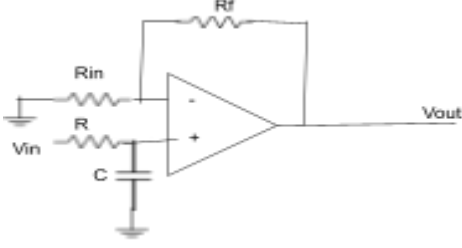
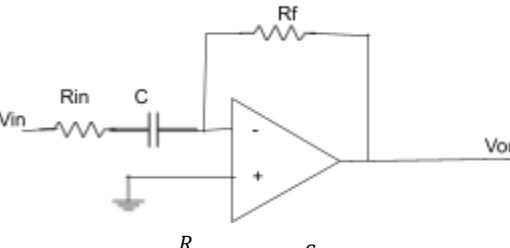
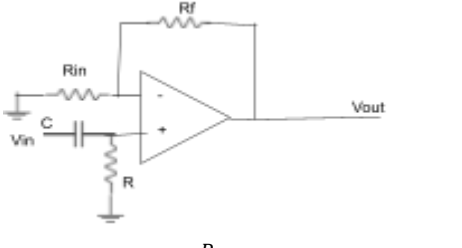
	Inverting	Non inverting
Gain:	$Av = -\frac{R_f}{R_{in}}$	$Av = 1 + \frac{R_f}{R_{in}}$
Low pass	 $H(s)_{active} = -\frac{R_f}{R_{in}} \times \frac{\frac{1}{RfC}}{S + \frac{1}{RfC}}$	 $H(s)_{active} = (1 + \frac{R_f}{R_{in}}) \times \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$
High pass	 $H(s)_{active} = -\frac{R_f}{R_{in}} \times \frac{S}{S + \frac{1}{RinC}}$	 $H(s)_{active} = (1 + \frac{R_f}{R_{in}}) \times \frac{S}{S + \frac{1}{RC}}$

Table 1. Transfer functions of active first order filters. In all cases the transfer function is the gain times the transfer function of the passive filter component, or $H(s)_{active} = AV \times H(s)_{passive}$

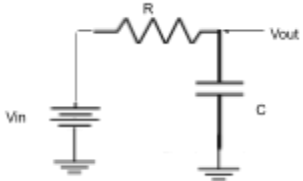
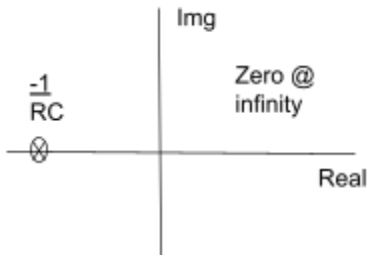
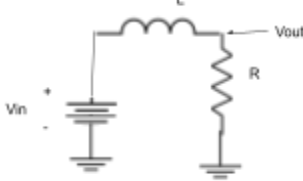
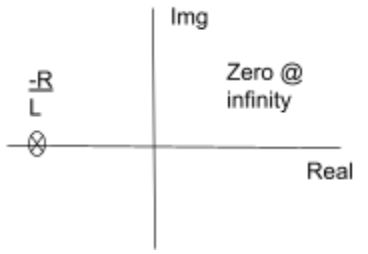
A shunt resistor is a resistor connected from a node to ground that provides a low-resistance path for current. This allows some of the current to flow to an alternative point in the circuit. In the inverting high-pass filter, the shunt resistor is labeled **R**, as

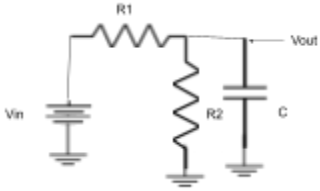
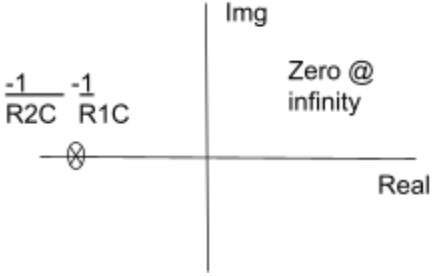
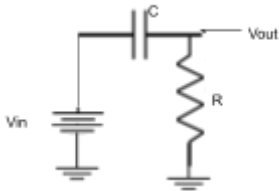
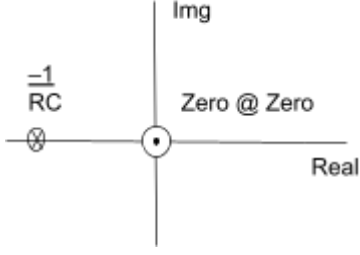
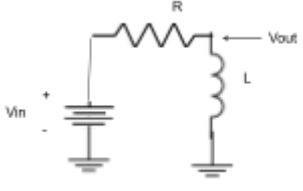
shown in Table 1. In contrast, the inverting low-pass filter and the non-inverting low-pass filter do not contain a shunt resistor in this configuration.

The process for first order active filters is similar compared to first order passive filters in that the transfer function of the passive RC network needs to be determined as well. To determine if a filter is low- or high-pass, analysis of the total active transfer function (which consists of the gain times the passive component transfer function) can be executed. The numerator of the transfer function is analyzed to find the number of zeros (and to determine if the circuit is low/high pass) and the denominator is analyzed to find the number of poles (and to determine stability). The numerator is set to zero for analysis purposes. If the numerator is a constant, it cannot equal zero and thus indicates a zero at infinity. Having a zero at infinity indicates that the transfer function is for a circuit that is a low-pass filter. If the numerator is a variable value, when the numerator is set to zero, the variable now equals zero. This means that there is one zero at zero and indicates that the circuit is a high-pass filter. To determine if a filter is stable, the denominator is analyzed by setting the polynomial expression to zero. If the variable equals a negative value, the system is considered to be stable. This process is illustrated in Tables 1. Additionally, by analysing what the transfer function approaches as s goes to infinity or 0, we can see the behavior of the filter - low pass, high pass, band pass, or band stop.

An important characteristic of filters is the -3dB cutoff frequency. The -3dB point represents the frequency at which the output power drops to one half of the input power. The cutoff frequency occurs at the value of ω where the magnitude of the transfer function decreases to $\frac{1}{\sqrt{2}}$ (or 0.707) of its maximum value. This can be found by substituting $s = j\omega$ and then finding the magnitude of the transfer function. Because the filter begins transitioning from the

passband (where signals pass with minimal attenuation) to the attenuation region (where signal amplitude decreases), the -3dB point is used to define the filter's usable bandwidth. For example, for a bandpass filter the f_c of the highpass filter (located first) minus the f_c of the lowpass filter would be the bandwidth.

Filter type	f_c	Configuration of passive devices	Transfer function	Stability analysis
Low pass	$f_c = \frac{1}{R\pi 2C}$		$H(S) = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$	$\frac{\frac{1}{RC}}{S + \frac{1}{RC}} = \frac{n(Z)}{n(P)}$, so $n(Z) = \frac{1}{RC}$ and $n(p) = S + \frac{1}{RC}$. Setting $n(Z) = 0$ gives $\frac{1}{RC} = 0$. Since C cannot be zero, the constant $\frac{1}{RC}$ cannot equal 0. Therefore, we have a zero at infinity and thus a low pass filter. Setting $n(p) = 0$ gives $S + \frac{1}{RC} = 0$ and $S = -\frac{1}{RC}$. We have one pole on the negative side of the s plane, so the system is stable. 
	$f_c = \frac{R}{2\pi L}$		$H(S) = \frac{\frac{R}{L}}{S + \frac{R}{L}}$	$\frac{\frac{R}{L}}{S + \frac{R}{L}} = \frac{n(Z)}{n(P)}$, so $n(Z) = \frac{R}{L}$ and $n(p) = S + \frac{R}{L}$. Setting $n(Z) = 0$ gives $\frac{R}{L} = 0$. Since L cannot be zero, the constant $\frac{R}{L}$ cannot equal 0. Therefore, we have a zero at infinity and thus a low pass filter. Setting $n(p) = 0$ gives $S + \frac{R}{L} = 0$ and $S = -\frac{R}{L}$. We have one pole on the negative side of the s plane, so the system is stable. 

	$f_c = \frac{1}{R\pi 2C}$		$H(s) = \frac{\frac{1}{c}}{s + \frac{1}{R2C} + \frac{1}{R1C}}$	$\frac{\frac{1}{c}}{s + \frac{1}{R2C} + \frac{1}{R1C}} = \frac{n(Z)}{n(P)}$, so $n(Z) = \frac{1}{c}$ and $n(p) = s + \frac{1}{R2C} + \frac{1}{R1C}$. When we set $n(Z) = 0$, $\frac{1}{c} = 0$. Since C cannot be zero, the constant $\frac{1}{c}$ cannot equal 0. Therefore, we have a zero at infinity and thus a low-pass filter. Setting $n(p) = 0$ gives us $s + \frac{1}{R2C} + \frac{1}{R1C} = 0$ or $s = -\frac{1}{R2C} - \frac{1}{R1C}$. We have one pole on the negative side of the s plane, so the system is stable 
High pass	$f_c = \frac{1}{R\pi 2C}$		$H(S) = \frac{S}{s + \frac{1}{RC}}$	$\frac{S}{s + \frac{1}{RC}} = \frac{n(Z)}{n(P)}$, so $n(Z) = S$ and $n(p) = s + \frac{1}{RC}$. When we set $n(Z) = 0$, $S = 0$. Since $S = 0$, there is one zero at zero, and therefore, we have a high-pass filter. Setting $n(p) = 0$ gives $s + \frac{1}{RC} = 0$ and $s = -\frac{1}{RC}$. We have one pole on the negative side of the s plane, so the system is stable. 
	$f_c = \frac{R}{2\pi L}$		$H(S) = \frac{S}{s + \frac{R}{L}}$	$\frac{S}{s + \frac{R}{L}} = \frac{n(Z)}{n(P)}$, so $n(Z) = S$ and $n(p) = s + \frac{R}{L}$. Setting $n(Z) = 0$ gives us $S = 0$. Since $S = 0$, there is one zero at zero, and therefore, we have a high-pass filter. When we set $n(p) = 0$, $s + \frac{R}{L} = 0$ and so $s = -\frac{R}{L}$. We have one pole on the negative side of the s plane, so the system is stable.

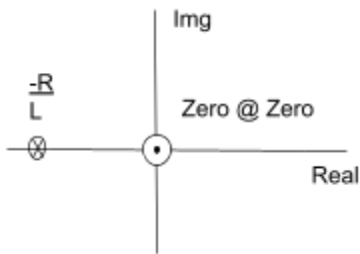
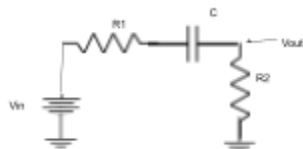
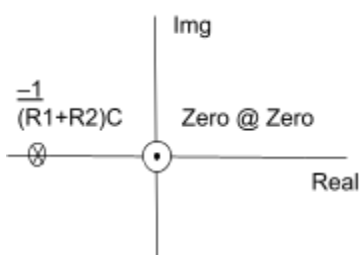
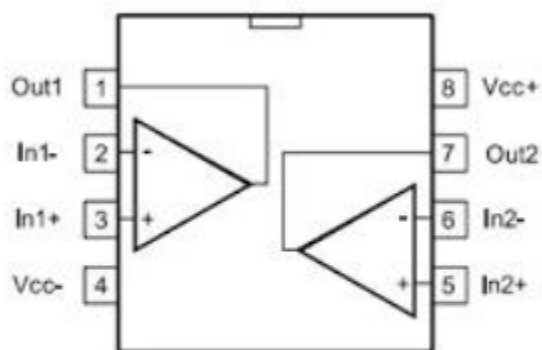
				
$f_c = \frac{1}{R\pi 2C}$		$H(s) = \frac{(\frac{R2}{R1+R2})}{S + \frac{1}{(R1+R2)C}}$	$\frac{(\frac{R2}{R1+R2})S}{S + \frac{1}{(R1+R2)C}} = \frac{n(Z)}{n(P)}$, so $n(Z) = (\frac{R2}{R1+R2})S$ and $n(p) = S + \frac{1}{(R1+R2)C}$. When we set $n(Z) = 0$, we still have $S = 0$. Therefore, we still have a zero at zero. Setting $n(p) = 0$ gives $S + \frac{1}{(R1+R2)C} = 0$ and therefore $S = -\frac{1}{(R1+R2)C}$. We have one pole on the negative side of the s plane, so the system is stable.	
				

Table 2. Transfer function and analysis for first order passive filters. This table can be referenced while finding the transfer function of active filters.

METHODOLOGY

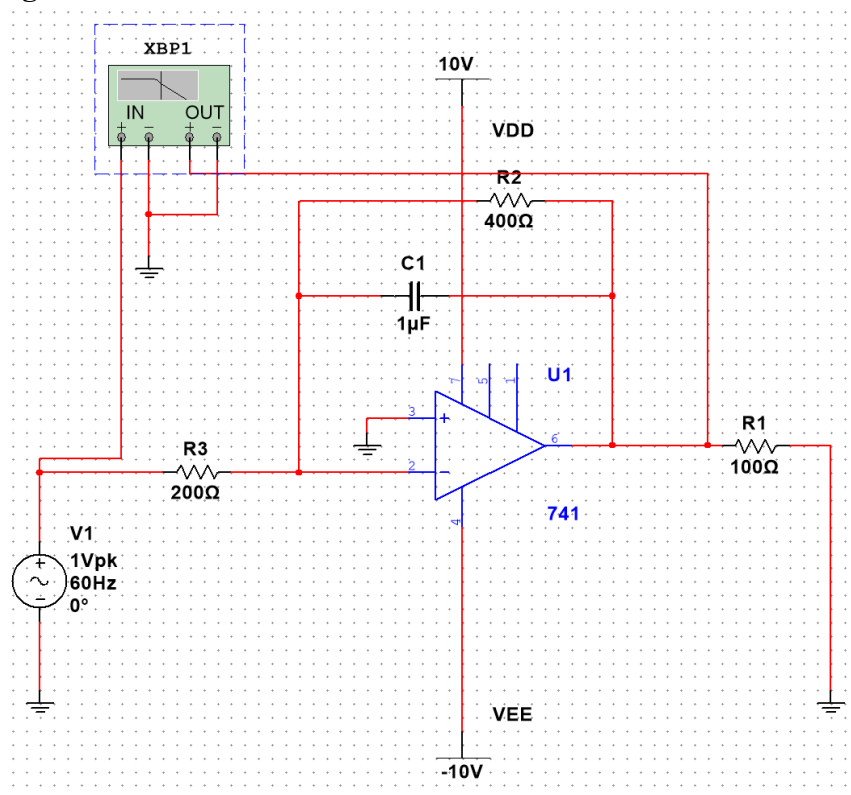
Each circuit was first designed theoretically by determining the required component values based on a desired cutoff frequency. In each circuit, the op amp LM358p was used. Then, with the chosen components the circuits were built on Multisim and the Bode plots generated. The bode plots were used to check the -3dB cutoff frequency and verify the predicted filter behavior. In each section for each circuit, transfer functions were derived for each circuit and analysis done to determine filter behavior was done based on these transfer functions. Additionally, the circuit for the active inverting low pass circuit as well as the active non

inverting high pass circuit were constructed on a breadboard and tested. Additionally, in each section the simulated and calculated results were compared to verify the transfer functions and confirm the expected filtering characteristics of each circuit.

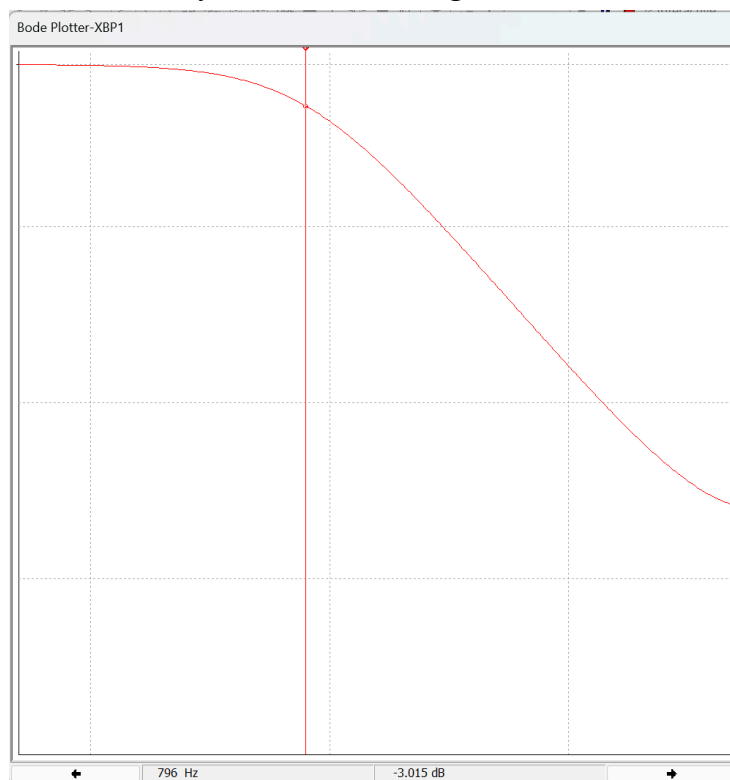


IMG 1. Pinout of the LM358P, from Jotrin.com. This was referenced while building the circuits.

Active Inverting LPF



IMG 6. Schematic from Multisim of an Active Inverting Low Pass Filter.



IMG 7. Bode of the simulated circuit, showing the -3dB cutoff frequency at 796Hz.

Calculations:

1. Finding component values based on critical frequency

$$XC = R, f_c = \frac{1}{2\pi RC}, R = \frac{1}{2\pi f_c C}$$

For a f_c of 796Hz and a chosen, ideal capacitor value of $1\mu F$

$$R = \frac{1}{2\pi(796)(1*10^{-6})} = 200\Omega.$$

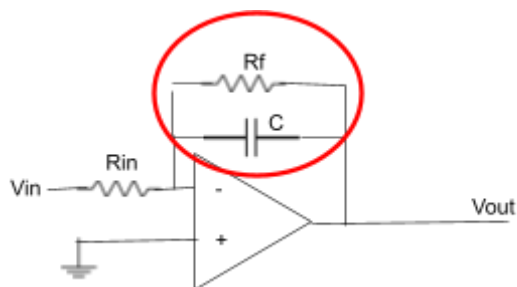
	Circuit from Multisim	Circuit from breadboard
R_{in}	200 Ω	200 Ω
R_f	400 Ω	400 Ω
C	1 μF	1 μF

Table 8. Materials used, along with the LM358p. Note that the resistor value we calculated is in row three, and is the resistor used in the passive filter section of the circuit.

2. Finding the gain:

Gain of an inverting op amp is $AV = -\frac{R_f}{R_{in}}$, which is $AV = -\frac{400}{200} = -2$

3. Find the passive transfer function:

**IMG 8.**

3a. Find V_{out} , based on image 8

General voltage divider formula: $V_{out} = V_{in} \frac{R2}{R2+R1}$

Voltage divider to find V_{out} : $V_{out} = V_{in} \frac{\frac{1}{CS}}{\frac{1}{CS}+R}, \quad \frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS}}{\frac{1}{CS}+R}$

3b. Clean up the transfer function with the goal of isolating S for stability analysis.

Normalize by multiplying by $\frac{1}{RC}$

$$\frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS}}{\frac{1}{CS} + R} \times \frac{\frac{1}{RC}}{\frac{1}{RC}} = \frac{\frac{1}{CS RC}}{\frac{1}{CS RC} + \frac{1}{C}}, \quad \frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS RC}}{\frac{1}{CS RC} + \frac{1}{C}} \times \frac{CS}{CS} = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

$$H(S) = \frac{V_{out}}{V_{in}}, \quad H(S) = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

4. The transfer function of an active filter is $H(s)_{active} = AV \times H(s)_{passive}$

$$H(s)_{active} = AV \times H(s)_{passive} = -\frac{R_f}{R_{in}} \times \frac{\frac{1}{RfC}}{S + \frac{1}{RfC}} = -2 \times \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

5. Stability and filter behavior analysis

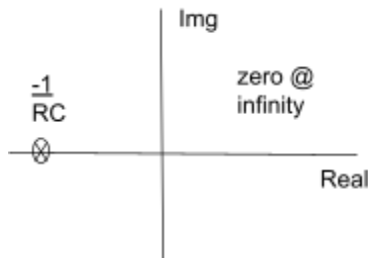
$$H(s)_{active} = \frac{\frac{-2}{RC}}{S + \frac{1}{RC}} = \frac{n(Z)}{n(P)}, \text{ so } n(Z) = \frac{-2}{RC} \text{ and } n(p) = S + \frac{1}{RC}$$

Setting $n(Z) = 0$ gives $\frac{-2}{RC} = 0$. Since C cannot be zero, the constant $\frac{5}{RC}$ cannot equal 0.

Therefore, we have a zero at infinity which indicates a **low-pass filter**. Setting $n(p) = 0$ gives

$S + \frac{1}{RC} = 0$ and $S = -\frac{1}{RC}$. We have one pole on the negative side of the s plane, which indicates

that the **system is stable**.

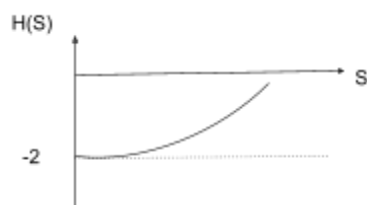


According to our analysis, we have a stable system with low pass filter behavior.

$$\text{Since } H(s)_{active} = \frac{\frac{-2}{RC}}{S + \frac{1}{RC}},$$

$$\text{As } S \rightarrow 0, \quad \frac{\frac{-2}{RC}}{S + \frac{1}{RC}} \approx -2 \times \frac{\frac{1}{RC}}{0 + \frac{1}{RC}} \approx -2 \times \frac{1}{1} \approx -2$$

$$\text{As } S \rightarrow \infty, \frac{\frac{-2}{RC}}{S + \frac{1}{RC}} \approx \frac{\frac{-2}{RC}}{\infty + \frac{1}{RC}} \approx -2 \times \frac{\frac{1}{RC}}{\infty} \approx -2 \times 0 \approx 0$$



Based on this analysis of the limits of s , the circuit passes low frequency and attenuates high frequency, thus **behaving like a low pass filter**.

6. Find $V_{out}(t)$ if $V_{in}(t) = 5u(t)$

Recall the Laplace transformations to convert $V_{in}(s)$

F(S)	F(T)
$\frac{1}{s}$	$u(t)$
$\frac{1}{s^2}$	$tu(t)$
$\frac{1}{s+\alpha}$	$e^{-\alpha t} u(t)$
$\frac{1}{s-\alpha}$	$e^{\alpha t} u(t)$
$\frac{s}{s^2+\omega^2}$	$\cos(\omega t) u(t)$
$\frac{\omega}{s^2+\omega^2}$	$\sin(\omega t) u(t)$

Table 9. Laplace transform table

Therefore $V_{in}(s) = 5 \frac{1}{s}$.

6a. Since $H(s) = \frac{V_{out}}{V_{in}}$, $V_{out}(s) = H(s)V_{in}(s)$,

$$V_{out}(s) = \frac{\frac{-2}{RC}}{S + \frac{1}{RC}} \times 5 \frac{1}{s} = \frac{-10}{S} \frac{1}{RC} \frac{1}{S + \frac{1}{RC}} = \frac{-10}{RC} \frac{1}{S(S + \frac{1}{RC})}$$

6b. Now, we need to prepare for la place by using partial fractions. The denominator in $\frac{-10}{S(S+\frac{1}{RC})}$

is not suitable for la place transformation, and we want $s + \alpha$. We can get it into a suitable form

through usage of partial fractions. Substitute $\tau = RC$, so $V_{out}(s) = \frac{-10}{\tau S(S+\frac{1}{\tau})} = \frac{-10}{S(S\tau+1)}$

Partial fractions: $\frac{-10}{S(S\tau+1)} = \frac{A}{S} + \frac{B}{(S\tau+1)}$

Clear denominator: $-10 = A(S\tau + 1) + B(S)$, expand: $-10 = AS\tau + A + BS$

Group and pull out S: $-10 = S(A\tau + B) + A$, then solve for A and B:

There is no S, so therefore it should be $-10 = S(0) + A$, so $A = -10$ and $A\tau + B = 0$. Therefore, $B =$

$-A\tau = 10\tau$. So, $V_{out}(s) = \frac{-10}{S} + \frac{10\tau}{(S\tau+1)}$. We fix the denominator in $\frac{10\tau}{(S\tau+1)}$, to isolate the S to

apply la place transforms : $\frac{10\tau}{(S\tau+1)} = \frac{10\tau}{\tau(S+\frac{1}{\tau})} = \frac{10}{(S+\frac{1}{\tau})}$

$$V_{out}(s) = \frac{-10}{S} + \frac{10}{(S+\frac{1}{\tau})}$$

Recall the la place transformations:

F(S)	F(T)
$\frac{1}{S}$	$u(t)$
$\frac{1}{S^2}$	$tu(t)$
$\frac{1}{S+\alpha}$	$e^{-\alpha t} u(t)$
$\frac{1}{S-\alpha}$	$e^{\alpha t} u(t)$
$\frac{S}{S^2+\omega^2}$	$\cos(\omega t) u(t)$
$\frac{\omega}{S^2+\omega^2}$	$\sin(\omega t) u(t)$

Table 10. Laplace transform table

$$\mathcal{L}^{-1}\left\{\frac{-10}{s}\right\} = -10u(t)$$

$$\mathcal{L}^{-1}\left\{\frac{10}{(s+\frac{1}{\tau})}\right\} = 10e^{-\frac{t}{\tau}}u(t) = 10e^{-\frac{t}{RC}}u(t)$$

$$\text{Therefore, } V_{out}(t) = -10(1 - e^{-\frac{t}{RC}})u(t)$$

6c. Plug in component values

$$\text{Since } R = 200\Omega \text{ and } C = 1\mu f, \tau = RC = 200\Omega \times 1 \times 10^{-6}F = 200\mu s$$

$$V_{out}(t) = -10(1 - e^{-\frac{t}{200\mu s}})u(t)$$

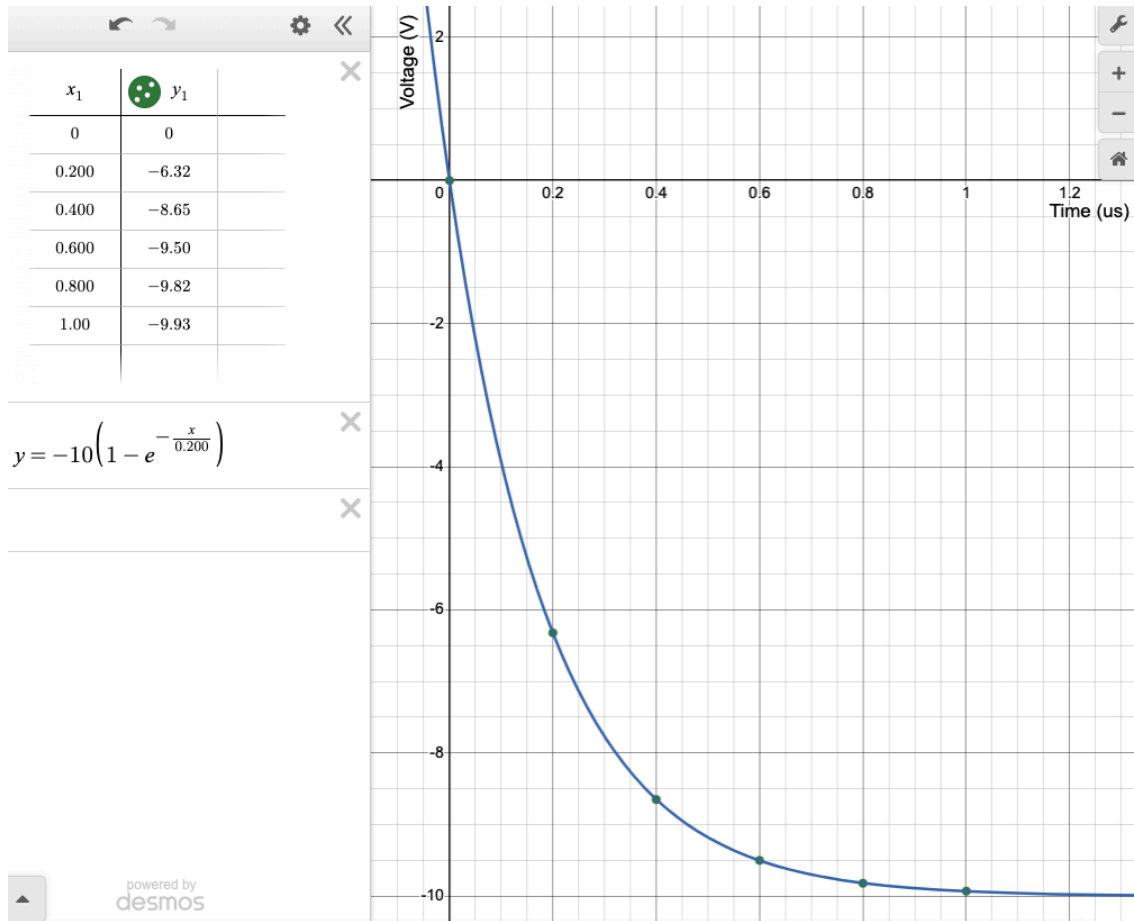
$$\text{For } 5\tau, 5\tau = 5(200\mu s) = 1ms. V_{out}(t) =$$

$$-10(1 - e^{-\frac{5t}{\tau}})u(t) = -10(1 - e^{-5}) = -9.93v$$

To find the % of the final value that the output is, $\frac{-9.93v}{-10v} * 100 = 99.33\%$

	$V_{out}(t)$	% Final
0τ	0v	0%
$1\tau = 200\mu s$	-6.32v	63.2%
$2\tau = 400\mu s$	-8.65v	86.5%
$3\tau = 600\mu s$	-9.50v	95.0%
$4\tau = 800\mu s$	-9.82v	98.2%
$5\tau = 1ms$	-9.93	99.3%

Table 11. $V_{out}(t)$ after time constants



IMG 9. Steady state is reached after 5τ , graphed using Desmos.com

5. We can also prove that the circuit is a low pass filter by calculating and applying the transfer function $H(\omega)$, with the goal of checking and describing the frequency response of the filter. This gives the theoretical method to analyze the filter behavior at different frequencies, including the current cutoff frequency of $F_c = 796\text{Hz}$, and attenuation of input signals.

Calculating $H(\omega)$:

$$H(S) = \frac{\frac{-2}{RC}}{S + \frac{1}{RC}} \text{ where } s = j\omega \text{ and the cutoff radian frequency is } \omega_c = \frac{1}{RC}$$

$$H(j\omega) = -2 \times \frac{\frac{1}{RC}}{s + \frac{1}{RC}} = -2 \times \frac{\omega_c}{j\omega + \omega_c}$$

$$\text{Simplifying } -2 \times \frac{\omega_c}{j\omega + \omega_c} \text{ gives } -2 \times \frac{\omega_c}{j\omega + \omega_c} \times \frac{\frac{1}{\omega_c}}{\frac{1}{\omega_c}} = -2 \times \frac{1}{1 + j\frac{\omega}{\omega_c}}$$

$$\text{If we choose } \omega = \omega_c, \text{ then } \frac{\omega}{\omega_c} = \frac{f}{f_c} \text{ and } -2 \times \frac{1}{1 + j\frac{\omega}{\omega_c}} = -2 \times \frac{1}{1 + j\frac{f}{f_c}}$$

Taking the magnitude gives us

$$|H(j\omega)| = \left| -2 \frac{1}{1 + j\frac{f}{f_c}} \right| = 2 \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}} = 2 \frac{1}{\sqrt{1 + \left(\frac{796}{796}\right)^2}} = 2 \frac{1}{\sqrt{1 + 1^2}} = 2 \frac{1}{\sqrt{2}} = 2 \times 0.707 = 1.41$$

The calculated magnitude is $20\log(1.41) = 2.98\text{dB}$.

The percent of error is calculated as

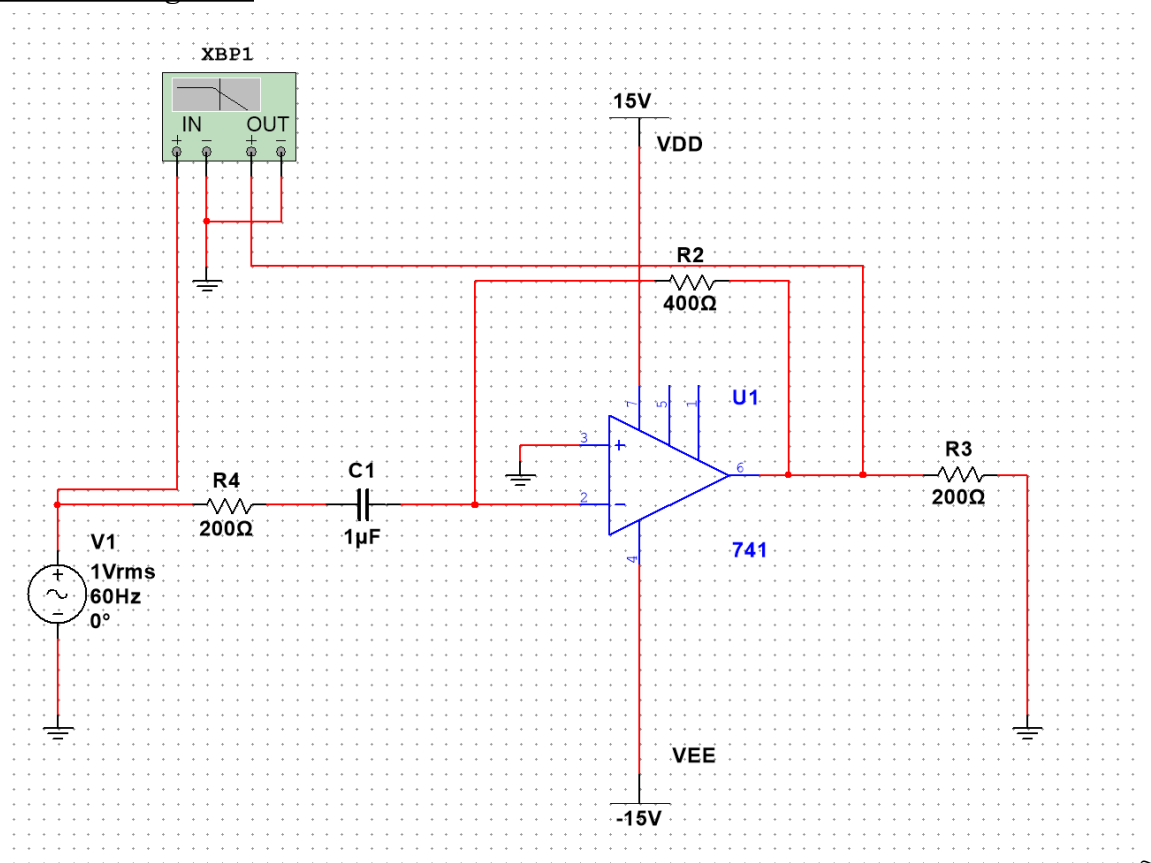
$$\frac{|\text{measured} - \text{calculated}|}{\text{calculated}} \times 100 = \frac{|0.783 - 0.707|}{0.707} \times 100 = 10.75\%$$

F_c calculated	F_c measured	Measured H(s)	Calculated H(s), NO GAIN	% of error of H(s)	Measured magnitude (dB)	Calculated magnitude (dB)
796Hz	796Hz	0.783	0.707	10.75%	-2.12	2.98

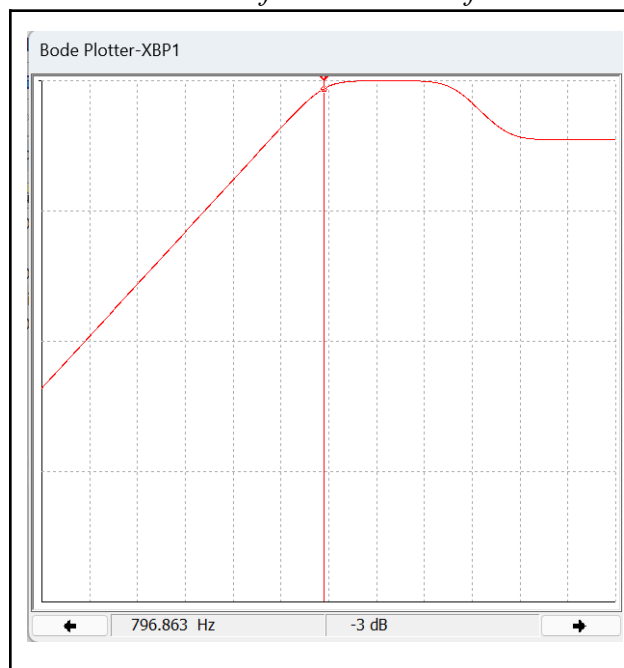
Table 12. RC measured verses calculated error

The calculated H(s) of the active inverting transfer function at the cutoff frequency is 1.41 or 2×0.707 . The calculated magnitude is $2 \times -3.01\text{ dB}$, or the measured magnitude was 0.783, or -2.12 dB . This results in a percent error of approximately 10.78%. The measured value being larger than the theoretical value indicates that the circuit attenuated the signal slightly less than predicted by the ideal transfer function. Despite this discrepancy, the measured response is still reasonably close to the theoretical value and confirms that the circuit exhibits the expected low-pass filter behavior with a cutoff near the calculated frequency.

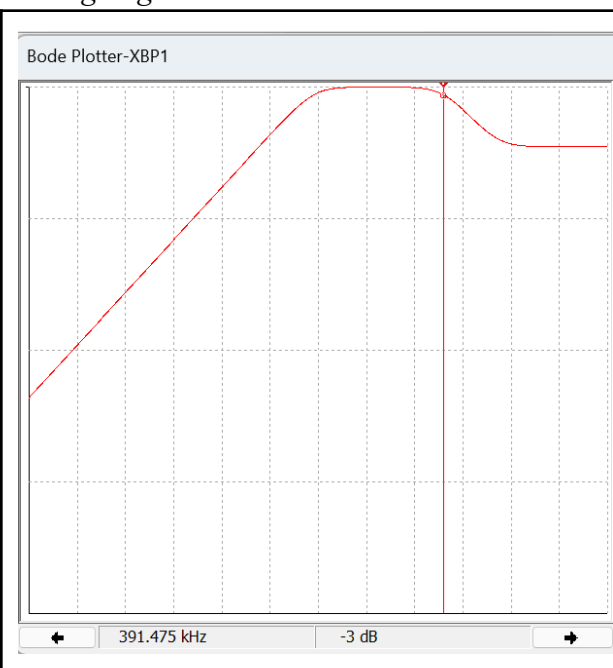
Active Inverting HPF



IMG 16. Schematic from Multisim of an Active Inverting High Pass Filter.



IMG 17. Bode of the simulated circuit, showing 3dB point at 796.863Hz



IMG 18. Bode plot of the simulated circuit, showing the 3dB point at 391.475kHz.

Calculations:

1. Finding component values based on critical frequency

$$XC = R, f_c = \frac{1}{2\pi RC}, R = \frac{1}{2\pi f_c C}$$

For a f_c of 796hz and a chosen, ideal capacitor value of $1\mu F$

$$R = \frac{1}{2\pi(796)(1*10^{-6})} = 199.94\Omega = 200\Omega.$$

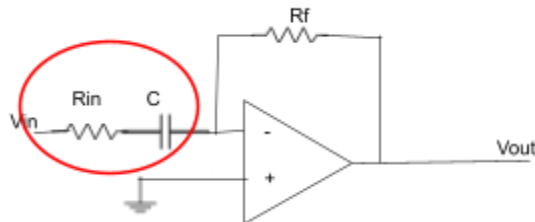
	Circuit from Multisim	Circuit from breadboard
R_{in}	200 Ω	120k Ω
R_f	400 Ω	400 Ω
R	200 Ω	200 Ω
C	1 μF	1 μF

Table 18. Materials used, along with the LM358p.

2. Finding the gain:

Gain of a non inverting op amp is $AV = -\frac{R_f}{R_{in}}$, which is $AV = -\frac{400}{200} = -2$

3. Find the passive transfer function:

**IMG 19.**

3a. Find V_{out} , based on image 14

$$V_{out} = V_{in} \frac{R}{R + \frac{1}{CS}}, \quad \frac{V_{out}}{V_{in}} = \frac{R}{R + \frac{1}{CS}}$$

Clean up the transfer function with the goal of isolating S for stability analysis, so

$$\text{multiply by CS: } H(s) = \frac{V_{out}}{V_{in}} = \frac{R}{R + \frac{1}{CS}} \times \frac{CS}{CS} = \frac{RCS}{RCS + 1}$$

$$H(s) = \frac{RCS}{RCS + 1} \times \frac{\frac{1}{RC}}{\frac{1}{RC}} = \frac{S}{S + \frac{1}{RC}}$$

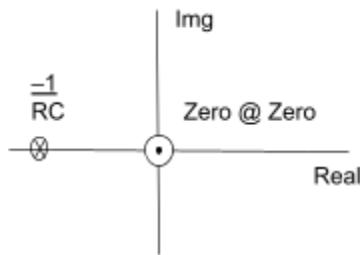
4. The transfer function of an active filter is $H(s)_{active} = AV \times H(s)_{passive}$

$$H(s)_{active} = AV \times H(s)_{passive} = -\frac{R_f}{R_{in}} \times \frac{S}{S + \frac{1}{RC}} = -2 \times \frac{S}{S + \frac{1}{RC}}$$

5. Stability and filter behavior analysis

$$H(s)_{active} = \frac{-2S}{S + \frac{1}{RC}} = \frac{n(Z)}{n(P)} \text{ so } n(Z) = -2S \text{ and } n(P) = S + \frac{1}{RC}$$

When we set $n(Z) = 0$, $-2S = 0$. Since $S = 0$, there is one zero at zero, and therefore we have a **high pass filter**. Setting $n(P) = 0$ gives $S + \frac{1}{RC} = 0$ and $S = -\frac{1}{RC}$. We have one pole on the negative side of the s plane, so **the system is stable**.



According to our analysis, we have a stable system with high pass filter behavior.

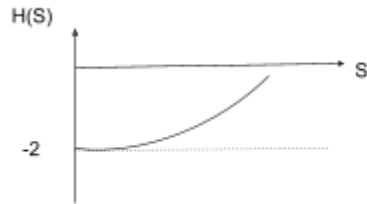
Since $H(s)_{active} = \frac{-2S}{S + \frac{1}{RC}}$

$$\text{As } S \rightarrow 0, \frac{-2S}{S + \frac{1}{RC}} \approx \frac{-2(0)}{(0) + \frac{1}{RC}} \approx 2.833 \times 0 \approx 0$$

$$\text{As } S \rightarrow \infty, \frac{-2S}{S + \frac{1}{RC}} \approx \frac{-2\infty}{\infty + \frac{1}{RC}} \approx -2 \times \frac{\infty}{\infty} \approx -2$$

$$\text{Applying L'Hopital for } \frac{\infty}{\infty} \text{ gives } \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} -2 \left(\frac{S}{S + \frac{1}{RC}} \right) = -2 \lim_{s \rightarrow \infty} \left(\frac{S}{S + \frac{1}{RC}} \right)$$

$$\left(\frac{\frac{d}{ds}(S)}{\frac{d}{ds}(S + \frac{1}{RC})} \right) = \left(\frac{1}{1+0} \right), \text{ so } -2 \lim_{s \rightarrow \infty} \left(\frac{1}{1} \right) = -2$$



Based on this analysis of the limits of s , the circuit passes low frequency and attenuates high frequency, thus **behaving like a inverting low pass filter**.

6. Find $V_{out}(t)$ if $V_{in}(t) = 5u(t)$

Recall the Laplace transformations to convert $V_{in}(s)$

F(S)	F(T)
$\frac{1}{s}$	$u(t)$
$\frac{1}{s^2}$	$tu(t)$
$\frac{1}{s+\alpha}$	$e^{-\alpha t} u(t)$
$\frac{1}{s-\alpha}$	$e^{\alpha t} u(t)$
$\frac{s}{s^2+\omega^2}$	$\cos(\omega t) u(t)$
$\frac{\omega}{s^2+\omega^2}$	$\sin(\omega t) u(t)$

Table 19. Laplace transform table

Therefore $V_{in}(s) = 5 \frac{1}{s}$.

6a. Since $H(s) = \frac{V_{out}}{V_{in}}$, $V_{out}(s) = H(s)V_{in}(s)$, and recall $H(s)_{active} = -2 \times \frac{s}{s + \frac{1}{RC}}$

$$V_{out}(s) = -2 \times \frac{s}{s + \frac{1}{RC}} \times 5 \frac{1}{s} = \frac{-10}{s} \frac{s}{s + \frac{1}{RC}} = \frac{-10}{s + \frac{1}{RC}}$$

6b. The denominator of $V_{out}(s) = \frac{-10}{s + \frac{1}{RC}}$ is suitable for Laplace transformation, so partial fractions are not needed.

Recall the Laplace transformations:

F(S)	F(T)
$\frac{1}{s}$	$u(t)$
$\frac{1}{s^2}$	$tu(t)$
$\frac{1}{s+\alpha}$	$e^{-\alpha t} u(t)$
$\frac{1}{s-\alpha}$	$e^{\alpha t} u(t)$
$\frac{s}{s^2 + \omega^2}$	$\cos(\omega t) u(t)$
$\frac{\omega}{s^2 + \omega^2}$	$\sin(\omega t) u(t)$

Table 20. Laplace transform table

$$V_{out}(s) = \frac{-10}{s + \frac{1}{RC}}$$

$$\mathcal{L}^{-1}\left\{\frac{-10}{(s + \frac{1}{\tau})}\right\} = -10e^{-\frac{t}{\tau}}u(t) = -10e^{-\frac{t}{RC}}u(t)$$

$$\text{Therefore, } V_{out}(t) = -10e^{-\frac{t}{RC}}u(t)$$

6c. Plug in component values

$$\text{Since } R = 200\Omega \text{ and } C = 1\mu f, \tau = RC = 200\Omega \times 1 \times 10^{-6}F = 200\mu s$$

$$V_{out}(t) = -10e^{-\frac{t}{200\mu}}u(t)$$

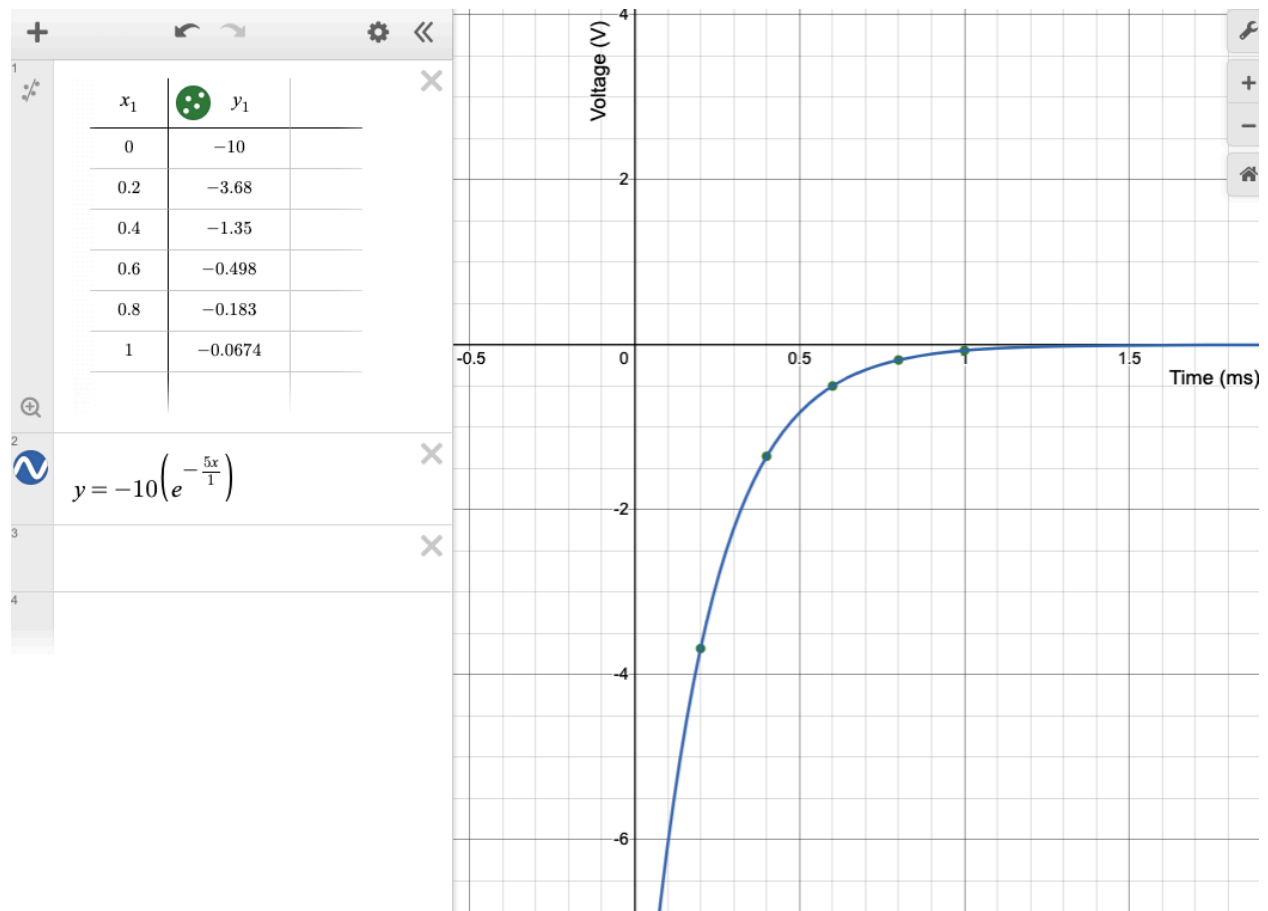
$$\text{For } 5\tau, 5\tau = 5(200\mu s) = 1ms. V_{out}(t) = -10(e^{-\frac{5t}{\tau}})u(t) = -10(e^{-5}) = 0.0674v$$

$$\% \text{ remaining} = \frac{0.0674}{10} * 100 = 0.674\%$$

% final change = 100% - % remaining = 100% - 0.674% = 99.3%

	$V_{out}(t)$	% Final
0τ	-10v	0%
$1\tau = 200\mu s$	-3.86v	63.2%
$2\tau = 400\mu s$	-1.35v	86.5%
$3\tau = 600\mu s$	-0.498	95.0%
$4\tau = 800\mu s$	-0.183	98.2%
$5\tau = 1ms$	-0.0674	99.3%

Table 21. $V_{out}(t)$ after time constants



IMG 15. Steady state is reached after 5τ , graphed using Desmos.com

5. We can also prove that the circuit is a low pass filter by calculating and applying the transfer function $H(\omega)$, with the goal of checking and describing the frequency response of the filter. This gives the theoretical method to analyze the filter behavior at different frequencies, including the current cutoff frequency of $F_c = 796\text{Hz}$, and attenuation of input signals.

Calculating $H(\omega)$:

$$H(S) = -2 \frac{S}{S + \frac{1}{RC}}, \text{ where } s = j\omega \text{ and } \omega = \frac{1}{RC},$$

$$\text{so } H(S) = -2 \frac{j\omega}{j\omega + \omega} = -2 \frac{j\omega}{\omega(j+1)} = -2 \frac{j}{(j+1)}$$

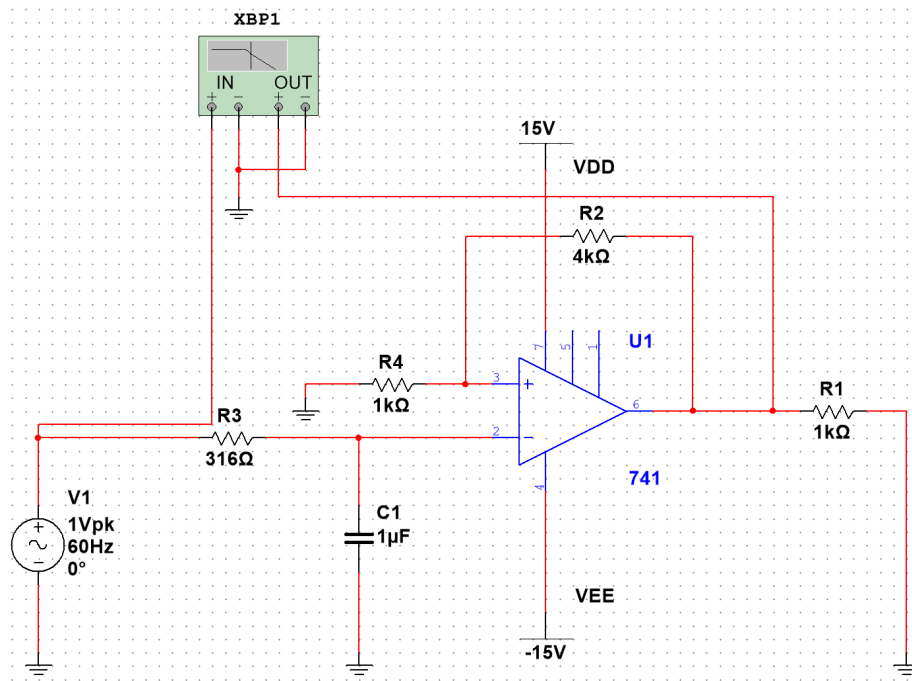
$$|H(j\omega)| = \left| -2 \times \frac{j}{1+j} \right| = 2 \frac{1}{\sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}} = 2 \times 0.707 = 1.414$$

F_c calculated	F_c measured	Measured H(s), NO GAIN	Calculated H(s), NO GAIN	% error of F(c)	Measured magnitude (dB) NO GAIN	Calculated magnitude (dB) NO GAIN
796	796.863	0.707	0.707	0.108%	-3.01	-3.01

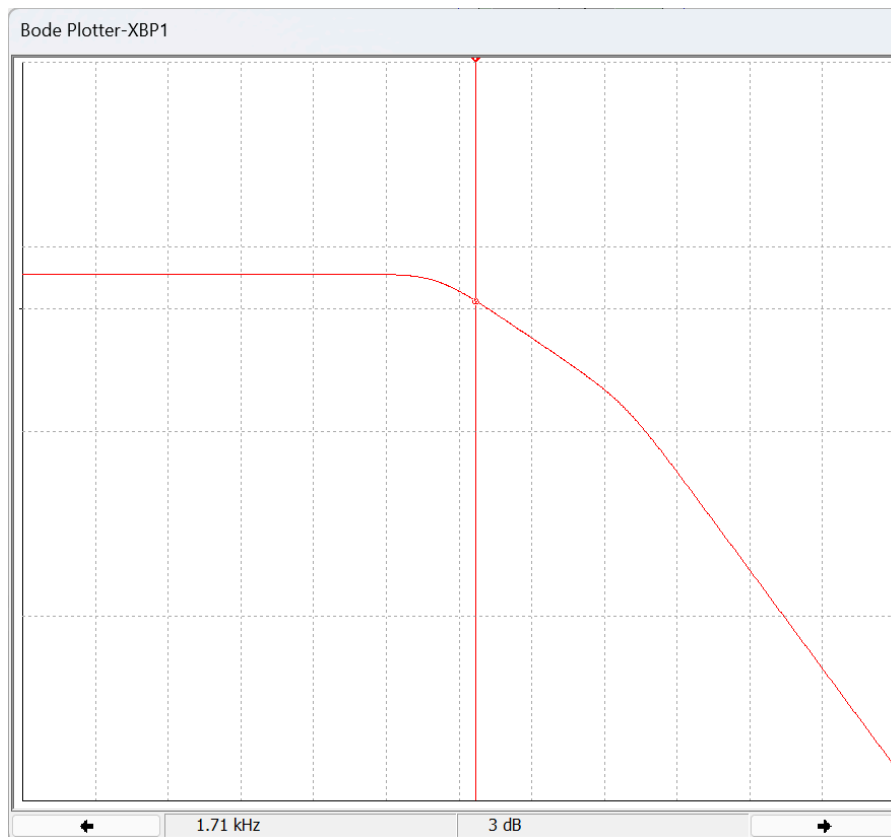
Table 22.

The calculated cutoff frequency of the active inverting high-pass filter was 796 Hz, while the simulated Bode plot showed a cutoff of 796.863 Hz. The normalized magnitude at the cutoff frequency is 0.707 (-3.01 dB), which matches the theoretical value for a first-order filter. The very small difference between measured and calculated results indicates strong agreement between theory and simulation.

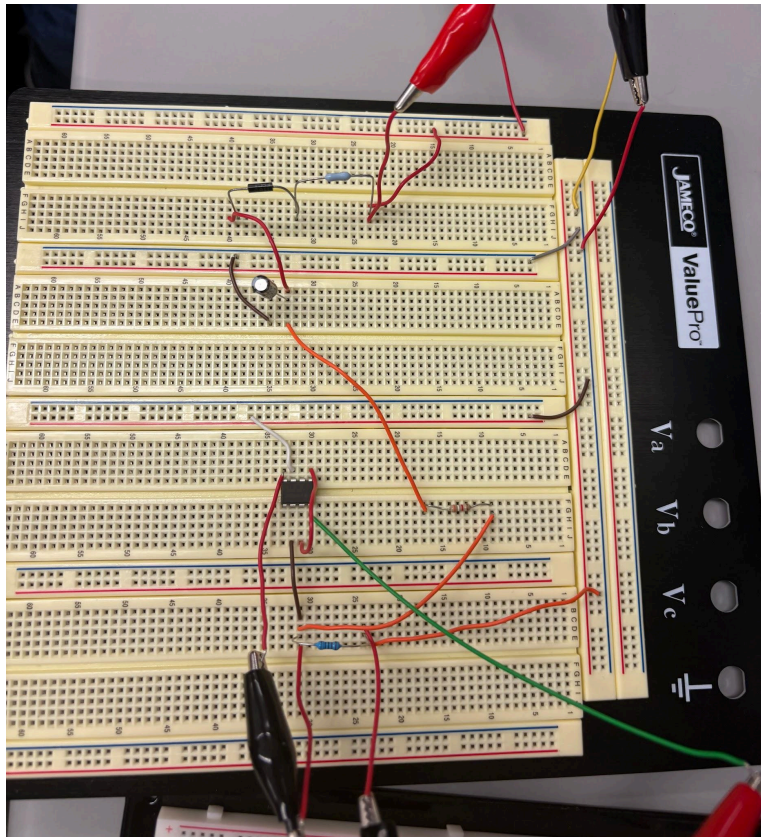
Active Non-Inverting Low Pass Filter



IMG 2. Schematic from Multisim of an active non-Inverting low pass filter.



IMG 3. Bode of the simulated circuit, showing the -3dB cutoff frequency at 1.71kHz.



IMG 4. Active non inverting low pass filter, physical circuit.

Calculations:

1. Finding component values based on critical frequency

$$XC = R, f_c = \frac{1}{2\pi RC}, R = \frac{1}{2\pi f_c C}$$

For a f_c of 504hz and a chosen, ideal capacitor value of $1\mu F$

$$R = \frac{1}{2\pi(504)(1 \times 10^{-6})} = 316\Omega.$$

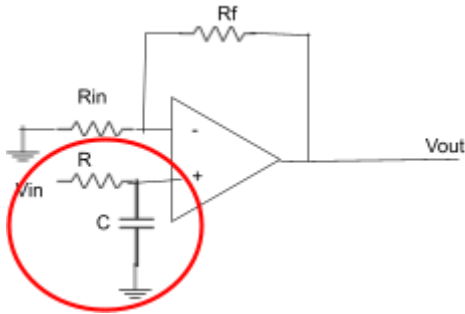
	Circuit from Multisim	Circuit from breadboard
R_{in}	1k Ω	1k Ω
R_f	4k Ω	4k Ω
R	316 Ω	118 Ω + 196 Ω = 314 Ω
C	1 μF	1 μF

Table 3. Materials used, along with the LM358p.

2. Finding the gain:

Gain of a non inverting op amp is $AV = 1 + \frac{R_f}{R_{in}}$, which is $AV = 1 + \frac{4k}{1k} = 5$

3. Find the passive transfer function:



IMG 4.

3a. Find V_{out} , based on image 4

General voltage divider formula: $V_{out} = V_{in} \frac{R_2}{R_2 + R_1}$

Voltage divider to find V_{out} : $V_{out} = V_{in} \frac{\frac{1}{CS}}{\frac{1}{CS} + R}$, $\frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS}}{\frac{1}{CS} + R}$

3b. Clean up the transfer function with the goal of isolating S for stability analysis.

Normalize by multiplying by $\frac{1}{RC}$

$$\frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS}}{\frac{1}{CS} + R} \times \frac{\frac{1}{RC}}{\frac{1}{RC}} = \frac{\frac{1}{CS RC}}{\frac{1}{CS RC} + \frac{1}{C}}, \quad \frac{V_{out}}{V_{in}} = \frac{\frac{1}{CS RC}}{\frac{1}{CS RC} + \frac{1}{C}} \times \frac{CS}{CS} = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

$$H(S) = \frac{V_{out}}{V_{in}}, \quad H(S) = \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

4. The transfer function of an active filter is $H(s)_{active} = AV \times H(s)_{passive}$

$$H(s)_{active} = AV \times H(s)_{passive} = \left(1 + \frac{R_f}{R_{in}}\right) \times \frac{\frac{1}{RC}}{S + \frac{1}{RC}} = 5 \times \frac{\frac{1}{RC}}{S + \frac{1}{RC}}$$

5. Stability and filter behavior analysis

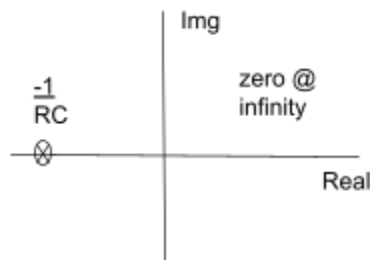
$$H(s)_{active} = \frac{\frac{5}{RC}}{S + \frac{1}{RC}} = \frac{n(Z)}{n(P)}, \text{ so } n(Z) = \frac{5}{RC} \text{ and } n(P) = S + \frac{1}{RC}$$

Setting $n(Z) = 0$ gives $\frac{5}{RC} = 0$. Since C cannot be zero, the constant $\frac{5}{RC}$ cannot equal 0.

Therefore, we have a zero at infinity which indicates **a low-pass filter**. Setting $n(p) = 0$ gives

$S + \frac{1}{RC} = 0$ and $S = -\frac{1}{RC}$. We have one pole on the negative side of the s plane, which indicates

that the **system is stable**.

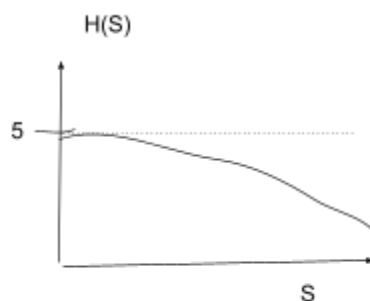


According to our analysis, we have a stable system with low pass filter behavior.

Since $H(s)_{active} = \frac{\frac{5}{RC}}{S + \frac{1}{RC}}$,

As $S \rightarrow 0$, $\frac{\frac{5}{RC}}{S + \frac{1}{RC}} \approx 5 \times \frac{\frac{1}{RC}}{0 + \frac{1}{RC}} \approx 5 \times \frac{1}{1} \approx 5$

As $S \rightarrow \infty$, $\frac{\frac{5}{RC}}{S + \frac{1}{RC}} \approx \frac{\frac{5}{RC}}{\infty + \frac{1}{RC}} \approx 5 \times \frac{\frac{1}{RC}}{\infty} \approx 5 \times 0 \approx 0$



Based on this analysis of the limits of s , the circuit passes low frequency and attenuates high frequency, thus **behaving like a low pass filter**.

6. Find $V_{out}(t)$ if $V_{in}(t)=5u(t)$

Recall the Laplace transformations to convert $V_{in}(s)$

F(S)	F(T)
$\frac{1}{s}$	$u(t)$
$\frac{1}{s^2}$	$tu(t)$
$\frac{1}{s+\alpha}$	$e^{-\alpha t}u(t)$
$\frac{1}{s-\alpha}$	$e^{\alpha t}u(t)$
$\frac{s}{s^2+\omega^2}$	$\cos(\omega t)u(t)$
$\frac{\omega}{s^2+\omega^2}$	$\sin(\omega t)u(t)$

Table 4. Laplace transform table

Therefore $V_{in}(s) = 5 \frac{1}{s}$.

6a. Since $H(s) = \frac{V_{out}}{V_{in}}$, $V_{out}(s) = H(s)V_{in}(s)$,

$$V_{out}(s) = \frac{\frac{5}{RC}}{s + \frac{1}{RC}} \times 5 \frac{1}{s} = \frac{25}{s} \frac{1}{RC} \frac{1}{s + \frac{1}{RC}} = \frac{25}{RC} \frac{1}{s(s + \frac{1}{RC})}$$

6b. Now, we need to prepare for Laplace by using partial fractions.

The denominator in $\frac{25}{s(s + \frac{1}{RC})}$ is not suitable for Laplace transformation, and we want $s + \alpha$. We

can get it into a suitable form through usage of partial fractions. Substitute $\tau = RC$, so

$$V_{out}(s) = \frac{25}{\tau s(s + \frac{1}{\tau})} = \frac{25}{s(s\tau + 1)}$$

$$\text{Partial fractions: } \frac{25}{s(s\tau + 1)} = \frac{A}{s} + \frac{B}{(s\tau + 1)}$$

Clear denominator: $25 = A(s\tau + 1) + B(s)$

Expand: $25 = AS\tau + A + BS$

Group and pull out S: $25 = S(A\tau + B) + A$

Solve for A and B:

There is no S, so therefore it should be $25 = S(0) + A$, so $A = 25$ and $A\tau + B = 0$. Therefore, $B = -A\tau = -25\tau$

$$\text{So, } V_{out}(s) = \frac{25}{s} + \frac{-25\tau}{(s\tau+1)}$$

We fix the denominator in $\frac{-25\tau}{(s\tau+1)}$, to isolate the S to apply Laplace transforms

$$\frac{-25\tau}{(s\tau+1)} = \frac{-25\tau}{\tau(s+\frac{1}{\tau})} = \frac{-25}{(s+\frac{1}{\tau})}$$

$$V_{out}(s) = \frac{25}{s} + \frac{-25}{(s+\frac{1}{\tau})}$$

Recall the Laplace transformations:

F(S)	F(T)
$\frac{1}{s}$	$u(t)$
$\frac{1}{s^2}$	$tu(t)$
$\frac{1}{s+\alpha}$	$e^{-\alpha t} u(t)$
$\frac{1}{s-\alpha}$	$e^{\alpha t} u(t)$
$\frac{s}{s^2+\omega^2}$	$\cos(\omega t) u(t)$
$\frac{\omega}{s^2+\omega^2}$	$\sin(\omega t) u(t)$

Table 5. Laplace transform table

$$\mathcal{L}^{-1}\left\{\frac{25}{s}\right\} = 25u(t)$$

$$\mathcal{L}^{-1}\left\{\frac{-25}{(s+\frac{1}{\tau})}\right\} = 25e^{-\frac{t}{\tau}}u(t) = 25e^{-\frac{t}{RC}}u(t)$$

Therefore , $V_{out}(t) = 25(1 - e^{-\frac{t}{RC}})u(t)$

6c. Plug in component values

Since $R = 316\Omega$ and $C = 1\mu f$, $\tau = RC = 316\Omega \times 1 \times 10^{-6}F = 316\mu s$

$$V_{out}(t) = 25(1 - e^{-\frac{t}{316\mu s}})u(t)$$

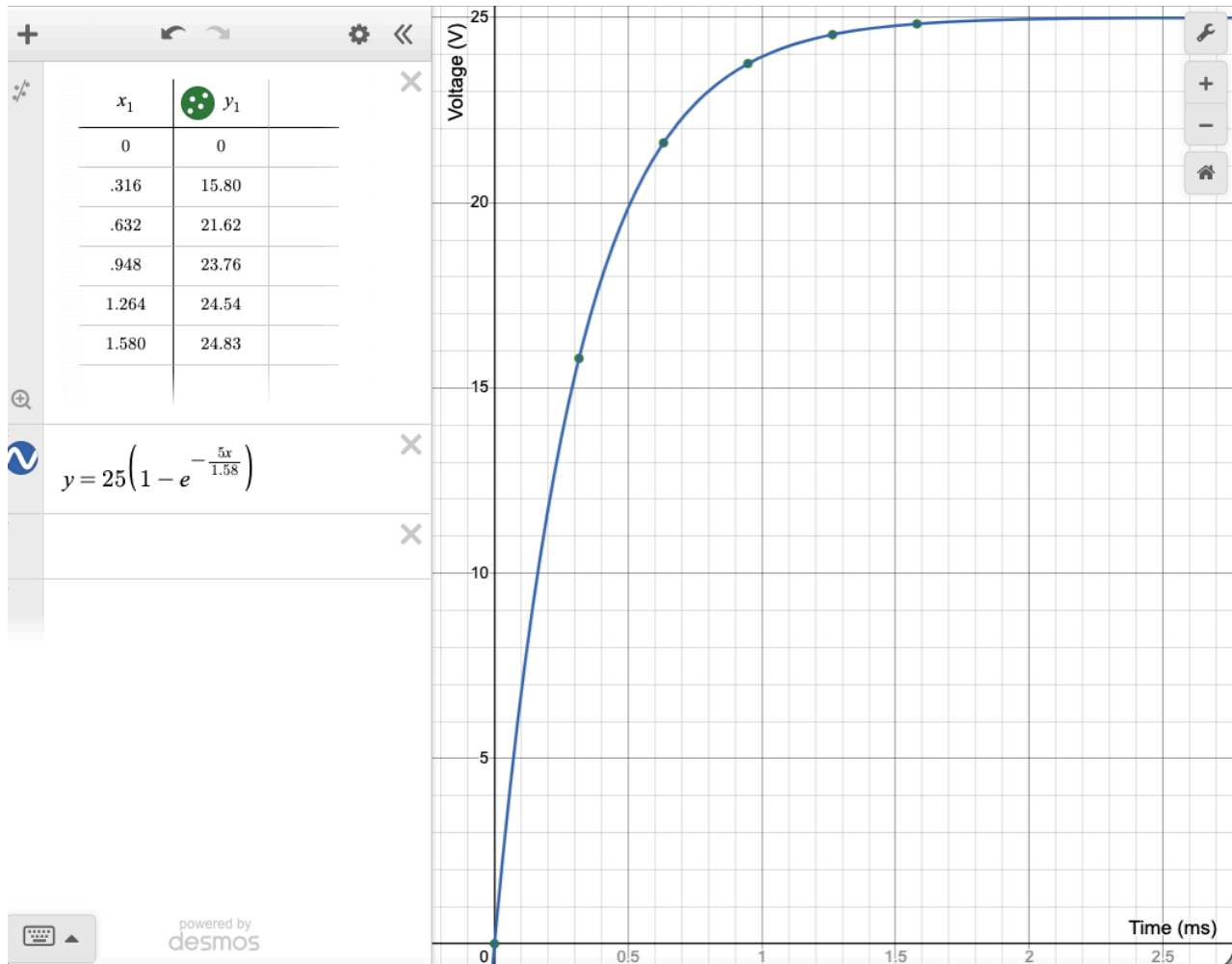
For 5τ , $5\tau = 5(316\mu s) = 1.58ms$. $V_{out}(t) =$

$$25(1 - e^{-\frac{5t}{\tau}})u(t) = 25(1 - e^{-5}) = 24.83v$$

To find the % of the final value that the output is, $\frac{24.83v}{25v} * 100 = 99.33\%$

	$V_{out}(t)$	% Final
0τ	0v	0%
$1\tau = 316\mu s$	15.8v	63.2%
$2\tau = 632\mu s$	21.6v	86.5%
$3\tau = 948\mu s$	23.8v	95.0%
$4\tau = 1.264ms$	24.5v	98.2%
$5\tau = 1.580ms$	24.83v	99.3%

Table 6. $V_{out}(t)$ after time constants



IMG 5. Steady state is reached after 5τ , graphed using [Desmos.com](https://www.desmos.com)

5. We can also prove that the circuit is a low pass filter by calculating and applying the transfer function $H(\omega)$, with the goal of checking and describing the frequency response of the filter. This gives the theoretical method to analyze the filter behavior at different frequencies, including the current cutoff frequency of $F_c = 504\text{Mhz}$, and attenuation of input signals.

Calculating $H(\omega)$:

$$H(S) = \frac{\frac{5}{RC}}{S + \frac{1}{RC}} \text{ where } s = j\omega \text{ and the cutoff radian frequency is } \omega_c = \frac{1}{RC}$$

$$H(j\omega) = 5 \times \frac{\frac{1}{RC}}{s + \frac{1}{RC}} = 5 \times \frac{\omega_c}{j\omega + \omega_c}$$

$$\text{Simplifying } 5 \times \frac{\omega_c}{j\omega + \omega_c} \text{ gives } 5 \times \frac{\omega_c}{j\omega + \omega_c} \times \frac{\frac{1}{\omega_c}}{\frac{1}{\omega_c}} = 5 \times \frac{1}{1 + j\frac{\omega}{\omega_c}}$$

$$\text{If we choose } \omega = \omega_c, \text{ then } \frac{\omega}{\omega_c} = \frac{f}{f_c} \text{ and } 5 \times \frac{1}{1 + j\frac{\omega}{\omega_c}} = 5 \times \frac{1}{1 + j\frac{f}{f_c}}$$

Taking the magnitude gives us

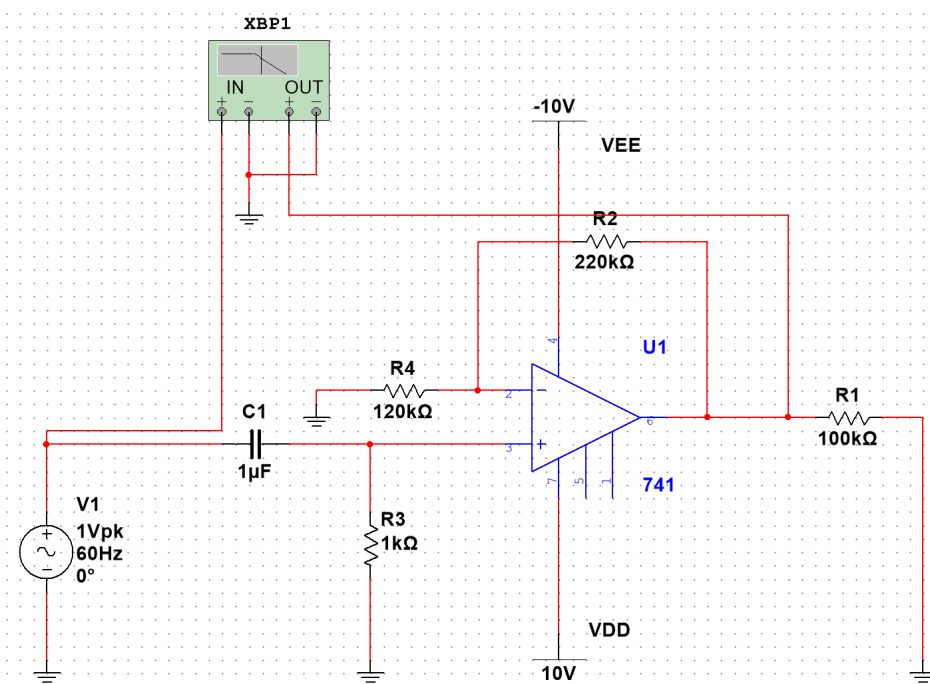
$$|H(j\omega)| = \left| 5 \frac{1}{1 + j\frac{f}{f_c}} \right| = 5 \frac{1}{\sqrt{1 + \frac{f^2}{f_c^2}}} = 5 \frac{1}{\sqrt{1 + \frac{504^2}{504^2}}} = 5 \frac{1}{\sqrt{1 + 1^2}} = 5 \frac{1}{\sqrt{2}} = 5 \times 0.707 = 3.536$$

F_c calculated	F_c measured	Measured $H(S)$	Calculated $H(S)$	% error of F_c	Measured magnitude (dB)	Calculated magnitude (dB)
503.8Hz	1.71kHz	0.707	3.535	239.4%	-3.01	10.97dB

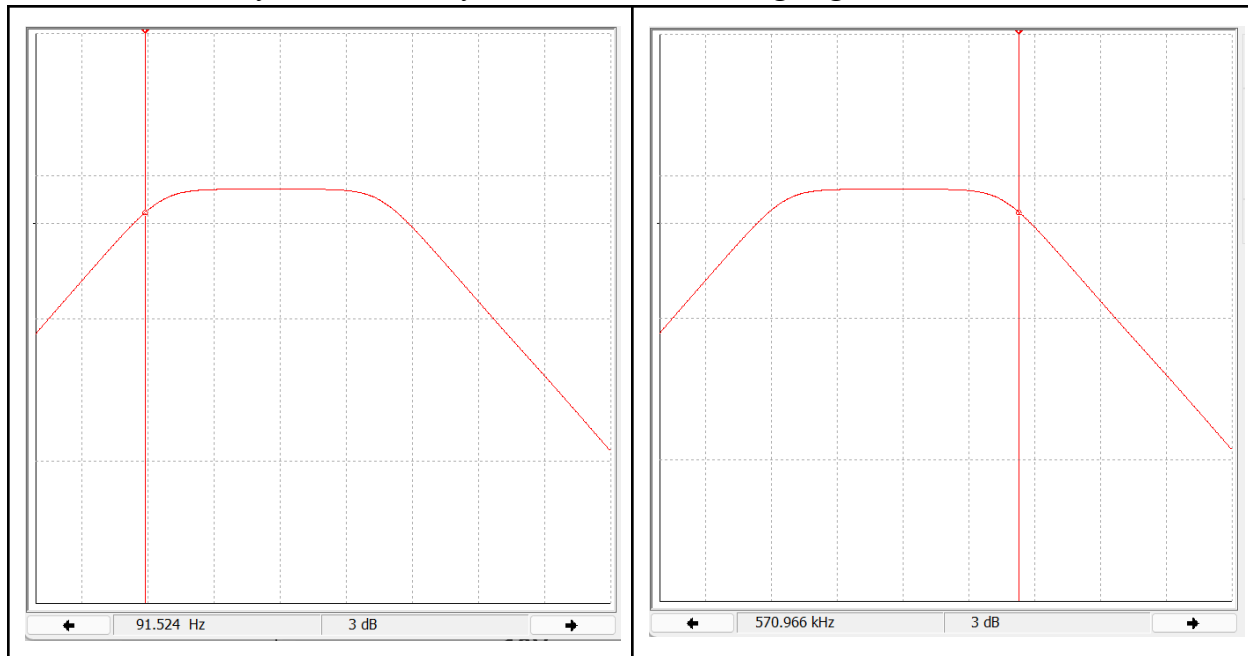
Table 7.

The calculated cutoff frequency for the active non-inverting low-pass filter was 503.8 Hz while the simulated Bode plot indicated a cutoff frequency of approximately 1.71 kHz. This corresponds to a percent error of approximately 239.4% between the theoretical and simulated cutoff frequencies. At the cutoff frequency, the normalized magnitude of a first-order filter is theoretically 0.707, which corresponds to -3.01 dB relative to the maximum response. Since the amplifier gain is 5 the theoretical magnitude at the cutoff frequency is $5 \times 0.707 = 3.535$, corresponding to approximately 10.97 dB, which is 3 dB below the passband gain of 13.98 dB. The simulated Bode plot confirms the -3 dB attenuation at the measured cutoff frequency. Unlike the other circuits analyzed in this report, this configuration shows a larger discrepancy between the theoretical and simulated cutoff frequencies.

Active non-inverting HPF

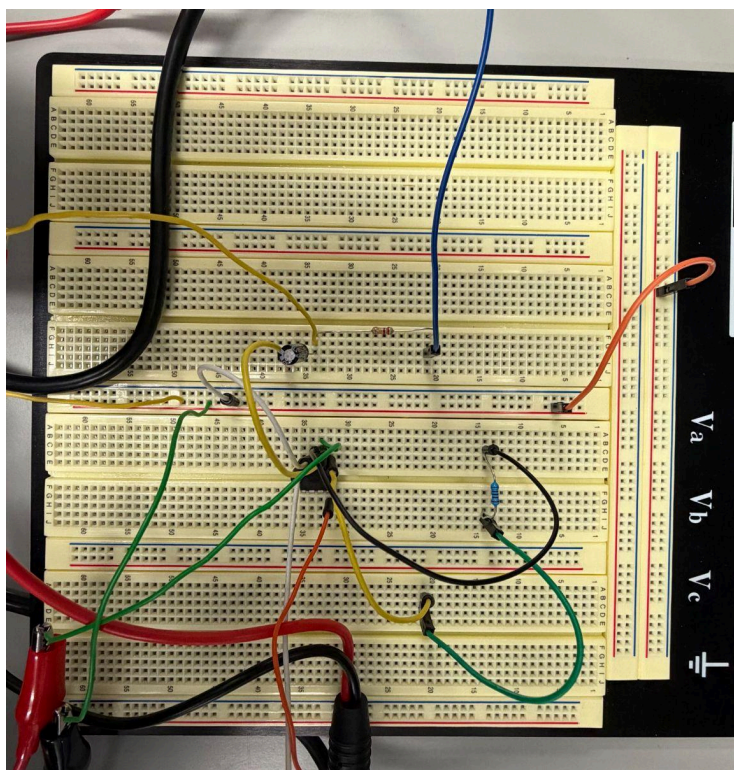


IMG 10. Schematic from Multisim of an Active Non Inverting High Pass Filter.



IMG 11. Bode of the simulated circuit, showing 3dB point at 91.524hz

IMG 12. Bode plot of the simulated circuit, showing the 3dB point at 570.966kHz.



IMG 13. Active Non Inverting High Pass Filter, built on a breadboard.

Calculations:

1. Finding component values based on critical frequency

$$XC = R, f_c = \frac{1}{2\pi RC}, R = \frac{1}{2\pi f_c C}$$

For a f_c of 91.52hz and a chosen, ideal capacitor value of $1\mu F$

$$R = \frac{1}{2\pi(91.52)(1 \cdot 10^{-6})} = 1k\Omega.$$

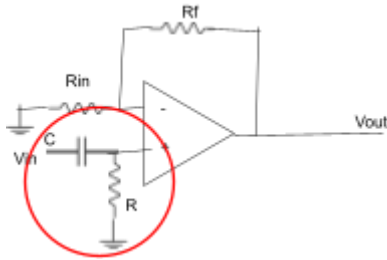
	Circuit from Multisim	Circuit from breadboard
R_{in}	120kΩ	120kΩ
R_f	220kΩ	220kΩ
R	1kΩ	1kΩ
C	1μF	1μF

Table 13. Materials used, along with the LM358p.

2. Finding the gain:

Gain of a non inverting op amp is $AV = 1 + \frac{R_f}{R_{in}}$, which is $AV = 1 + \frac{220k}{120k} = 2.833$

3. Find the passive transfer function:



IMG 14.

3a. Find V_{out} , based on image 14

$$V_{out} = V_{in} \frac{R}{R + \frac{1}{CS}}, \quad \frac{V_{out}}{V_{in}} = \frac{R}{R + \frac{1}{CS}}$$

Clean up the transfer function with the goal of isolating S for stability analysis, so

multiply by CS: $H(s) = \frac{V_{out}}{V_{in}} = \frac{R}{R + \frac{1}{CS}} \times \frac{CS}{CS} = \frac{RCS}{RCS + 1}$

$$H(s) = \frac{RCS}{RCS + 1} \times \frac{\frac{1}{RC}}{\frac{1}{RC}} = \frac{S}{S + \frac{1}{RC}}$$

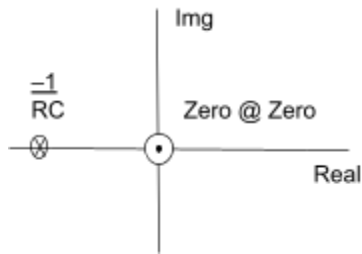
4. The transfer function of an active filter is $H(s)_{active} = AV \times H(s)_{passive}$

$$H(s)_{active} = AV \times H(s)_{passive} = \left(1 + \frac{R_f}{R_{in}}\right) \times \frac{S}{S + \frac{1}{RC}} = 2.833 \times \frac{S}{S + \frac{1}{RC}}$$

5. Stability and filter behavior analysis

$$H(s)_{active} = 2.833 \times \frac{S}{S + \frac{1}{RC}} = \frac{n(Z)}{n(P)} \text{ so } n(Z) = 2.833S \text{ and } n(P) = S + \frac{1}{RC}$$

When we set $n(Z) = 0$, $2.833S = 0$. Since $S = 0$, there is one zero at zero, and therefore we have a **high pass filter**. Setting $n(P) = 0$ gives $S + \frac{1}{RC} = 0$ and $S = -\frac{1}{RC}$. We have one pole on the negative side of the s plane, so **the system is stable**.



According to our analysis, we have a stable system with high pass filter behavior.

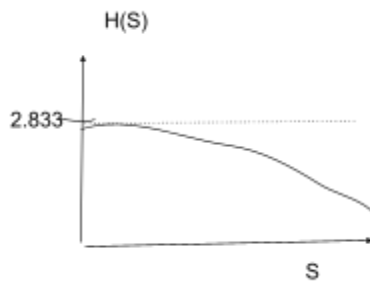
$$\text{Since } H(s)_{\text{active}} = 2.833 \times \frac{S}{S + \frac{1}{RC}},$$

$$\text{As } S \rightarrow 0, 2.833 \times \frac{S}{S + \frac{1}{RC}} \approx 2.833 \times \frac{(0)}{(0) + \frac{1}{RC}} \approx 2.833 \times 0 \approx 0$$

$$\text{As } S \rightarrow \infty, 2.833 \times \frac{S}{S + \frac{1}{RC}} \approx 2.833 \times \frac{(\infty)}{(\infty) + \frac{1}{RC}} \approx 2.833 \times \frac{\infty}{\infty} \approx 2.833$$

$$\text{Applying L'Hopital for } \frac{\infty}{\infty} \text{ gives } \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} 2.833 \left(\frac{S}{S + \frac{1}{RC}} \right) = 2.833 \lim_{s \rightarrow \infty} \left(\frac{S}{S + \frac{1}{RC}} \right)$$

$$\left(\frac{\frac{d}{ds}(S)}{\frac{d}{ds}(S + \frac{1}{RC})} \right) = \left(\frac{1}{1+0} \right), \text{ so } 2.833 \lim_{s \rightarrow \infty} \left(\frac{1}{1} \right) = 2.833$$



Based on this analysis of the limits of s , the circuit passes low frequency and attenuates high frequency, thus **behaving like a high pass filter**.

6. Find $V_{out}(t)$ if $V_{in}(t) = 5u(t)$

Recall the Laplace transformations to convert $V_{in}(s)$

F(S)	F(T)
$\frac{1}{s}$	$u(t)$
$\frac{1}{s^2}$	$tu(t)$
$\frac{1}{s+\alpha}$	$e^{-\alpha t} u(t)$
$\frac{1}{s-\alpha}$	$e^{\alpha t} u(t)$
$\frac{s}{s^2+\omega^2}$	$\cos(\omega t) u(t)$
$\frac{\omega}{s^2+\omega^2}$	$\sin(\omega t) u(t)$

Table 14. Laplace transform table

Therefore $V_{in}(s) = 5 \frac{1}{s}$.

6a. Since $H(s) = \frac{V_{out}}{V_{in}}$, $V_{out}(s) = H(s)V_{in}(s)$, and recall $H(s)_{active} = 2.833 \times \frac{s}{s+\frac{1}{RC}}$

$$V_{out}(s) = 2.833 \times \frac{s}{s+\frac{1}{RC}} \times 5 \frac{1}{s} = \frac{14.167}{s} \frac{s}{s+\frac{1}{RC}} = \frac{14.167}{s+\frac{1}{RC}}$$

6b. The denominator of $V_{out}(s) = \frac{14.167}{s+\frac{1}{RC}}$ is suitable for Laplace transformation, so partial

fractions are not needed.

Recall the Laplace transformations:

F(S)	F(T)
$\frac{1}{s}$	$u(t)$
$\frac{1}{s^2}$	$tu(t)$
$\frac{1}{s+\alpha}$	$e^{-\alpha t} u(t)$
$\frac{1}{s-\alpha}$	$e^{\alpha t} u(t)$

$\frac{s}{s^2 + \omega^2}$	$\cos(\omega t) u(t)$
$\frac{\omega}{s^2 + \omega^2}$	$\sin(\omega t) u(t)$

Table 15. Laplace transform table

$$V_{out}(s) = \frac{14.167}{s + \frac{1}{RC}}$$

$$\mathcal{L}^{-1}\left\{\frac{14.167}{(s + \frac{1}{\tau})}\right\} = 14.167e^{-\frac{t}{\tau}}u(t) = 14.167e^{-\frac{t}{RC}}u(t)$$

Therefore, $V_{out}(t) = 14.167e^{-\frac{t}{RC}}u(t)$

6c. Plug in component values

Since $R = 1k\Omega$ and $C = 1\mu f$, $\tau = RC = 1k\Omega \times 1 \times 10^{-6}F = 1000\mu s = 1ms$

$$V_{out}(t) = 14.167e^{-\frac{t}{1ms}}u(t)$$

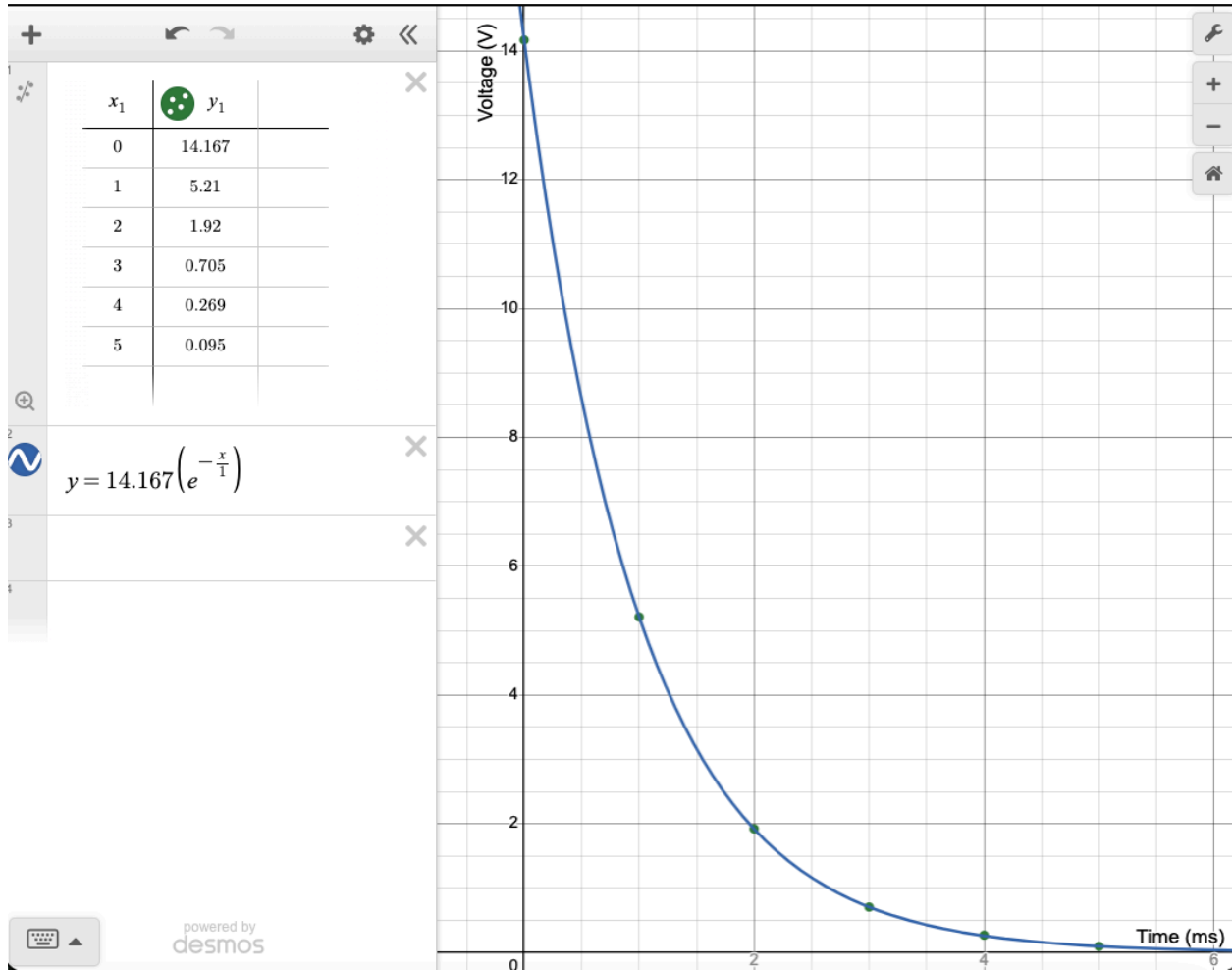
For 5τ , $5\tau = 5(1ms) = 5ms$. $V_{out}(t) = 14.167(e^{-\frac{5t}{\tau}})u(t) = 14.167(e^{-5}) = 0.095v$

$$\% \text{ remaining} = \frac{0.095}{14.167} * 100 = 0.674\%$$

$$\% \text{ final change} = 100\% - \% \text{ remaining} = 100\% - 0.674\% = 99.3\%$$

	$V_{out}(t)$	% Final
0τ	14.167v	0%
$1\tau = 1ms$	5.21v	63.2%
$2\tau = 2ms$	1.92v	86.5%
$3\tau = 3ms$	0.705v	95.0%
$4\tau = 4ms$	0.259v	98.2%
$5\tau = 5ms$	0.0954v	99.3%

Table 16. $V_{out}(t)$ after time constants



IMG 15. Steady state is reached after 5τ , graphed using Desmos.com

5. We can also prove that the circuit is a low pass filter by calculating and applying the transfer function $H(\omega)$, with the goal of checking and describing the frequency response of the filter. This gives the theoretical method to analyze the filter behavior at different frequencies, including the current cutoff frequency of $F_c = 159\text{Mhz}$, and attenuation of input signals.

Calculating $H(\omega)$:

$$H(S) = 2.833 \frac{S}{S + \frac{1}{RC}}, \text{ where } s = j\omega \text{ and } \omega = \frac{1}{RC},$$

$$\text{so } H(S) = 2.833 \frac{j\omega}{j\omega + \omega} = 2.833 \frac{j\omega}{\omega(j+1)} = 2.833 \frac{j}{(j+1)}$$

$$|H(j\omega)| = \left| 2.833 \times \frac{j}{1+j} \right| = 2.833 \frac{1}{\sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}} = 2.833 \times 0.707 = 2.003$$

The percent of error is calculated as

$$\frac{|f_c \text{ measured} - f_c \text{ calculated}|}{f_c \text{ calculated}} \times 100 = \frac{|91.524 - 91.52|}{91.52} \times 100 = 0.00437\%$$

F_c calculated	F_c measured	Measured H(s), NO GAIN	Calculated H(s), NO GAIN	% Error of fc	Measured magnitude (dB) NO GAIN	Calculated magnitude (dB) NO GAIN
91.52	91.524	0.707	0.707	0.00437%	-3.01	-3.01

Table 17.

The calculated cutoff frequency for the active non-inverting high-pass filter was 91.52Hz and the simulated Bode plot showed a measured cutoff frequency of approximately 91.524Hz. This corresponds to a percent error of approximately 0.0044% between the theoretical and simulated cutoff frequencies. At the cutoff frequency the magnitude of a first-order filter is 0.707, which corresponds to -3.01 dB. The Bode plot cursor was placed at the -3 dB point confirming that the simulated magnitude matches the theoretical value expected at the cutoff frequency. The extremely small difference between the calculated and simulated cutoff frequencies indicates very accurate agreement between the theoretical design equations and the simulated circuit behavior.

Discussion and conclusion

The purpose of this laboratory experiment was to design, simulate and analyze several first-order active filter circuits using operational amplifiers and RC networks. The experimental results closely matched the theoretical predictions derived from the transfer functions and cutoff frequency calculations. Analysis of the calculated transfer functions resulted in confirmation of expected behavior for frequency pass and attenuation, as well confirmed stability. In the case of the calculated $H(s)$ of the active inverting transfer function, the calculated and measured critical frequencies were both 796Hz, and the measured magnitude was 2.12 while the calculated was 2.98. The measured response was reasonably close with a 10.75% error of $H(s)$. For the active inverting high pass filter, the measured critical frequency was 796.863Hz while the calculated was 796, at the -3dB point. The analysis of the results confirmed very strong agreement between theory and simulation, with a 0.108% error. Unfortunately, the active non inverting low pass filter had a larger discrepancy between theoretical and simulated data, with a 239.4% error between calculated and measured cutoff frequencies. The theoretical and simulated data were in agreement for the active non inverting high pass filter, as there was a 0.00437% error of cutoff frequencies between the measured and calculated.

The overall agreement between theoretical, simulated, and measured results demonstrates that the designed and breadboard built circuits successfully function as first-order active filters, and that the RC design equations accurately predicted the filter behavior. The experiment also confirmed the expected behavior of active filter circuits and demonstrated how combining passive RC networks with operational amplifiers allows control over both gain and frequency response. The results validate the theoretical transfer functions and illustrate how active filters can be effectively designed and analyzed for signal processing applications.

References:

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